0

0.1 REQUIREMENT LOG

0.1.1 List of requirements

Cameras Power Switch Channel Status	3
Cameras Power Switch Enable Status	3
Cameras Power Switch Output Current	2
Cameras Power Switch Output Voltage	3
Cameras Power Switch Register data	3
CFM - AD&ADAS configuration handling	7
CFM - Camera Power Control	18
CFM - Camera recovery publication interface	26
CFM - Consumers of Fault propagation	16
CFM - Detection of fault for open circuit	14
CFM - Detection of fault for short to battery	13
CFM - Detection of fault for short to ground	13
CFM - Detection of fault for Thermal shutdown	14
CFM - Detection of fault for Undervoltage/Overvoltage Lockout	16
CFM - DID Camera Part Number	19
CFM - DID Camera Power Switch Output Current	17
CFM - DID Camera Serial Number	19
CFM - DID Cameras Power Switch Enable Status	18
CFM - DID Cameras Power Switch Output Status	17
CFM - DID Cameras Power Switch Output Voltage	17
CFM - Fault de-maturation process	9
CFM - Fault during Initialization	10
CFM - Fault information propagation	14
CFM - Fault inputs towards CPDD for camera power cut-off	15
CFM - Fault maturation process	7
CFM - Fault on Camera Over Temperature	11
CFM - Fault Over voltage	12
CFM - Freeze Frame data	12
CFM - ISET fault monitor	13
CFM - Requesting DTC towards Diagnostic Kernel	18
CFM - Un recoverable Fault during Runtime	10
CFM- Recoverable Fault during Run time	11
CFM: Internal Fault on the Deserializer	9
CPDD - AD&ADAS configuration handling	2
CPDD - Camera Safety Switch	4
CPDD - Hardware faults implementation	4
CPDD - I2C communication faults monitoring	2
CPDD Interval-based Monitoring periodicity	2
Enabling support for Hostlog and ZVS with DCF for Exterior Camera Perception data	16

0.1.2 New requirements in this issue

0.1.3 Updated requirements in this issue

CFM - Fault inputs towards CPDD for camera power cut-off	15
CFM - Fault maturation process	7
CPDD - Hardware faults implementation	4

0.1.4 Deleted requirements in this issue

0.2 Document References

0.3 Abbreviations

0.4 Definitions

1 Camera Diagnostic Monitor [A] 1.2: Internal requirements

1.1 PC ECDP - Towards SW Element Camera Power Diagnostics Driver (CPDD)

REQ-461325v2 CPDD - AD&ADAS configuration handling

Purpose: To ensure that camera diagnostics are implemented on the correct camera channels across configurations.

The Software Element Camera Power Device Driver (CPDD) in Software Component Camera Diagnostic Monitor (CDM) shall only implement diagnostics for the camera channels populated for the active vehicle configuration.

REQ-339734v5 CPDD - I2C communication faults monitoring

General comments: Purpose
Detect communication failure with the MAX20087 Power chip 1 and 2.

Preconditions
Camera channel/MAX20087 Chip in question is powered.

Purpose: The purpose of this requirement is to ensure that the Camera Power Device Driver (CPDD) continuously monitors I2C communication to the MAX20087 power switch chips. By tracking and timestamping successful I2C transactions, CPDD enables effective diagnostic analysis and fault tracing for the camera system. This will also ensure timely identification of communication faults that can lead to camera performance issues.

Verification Method: Unit testing, integration testing

SW Element Camera Power Driver Diagnostics (CPDD) shall monitor and provide timestamp for I2C reading communication of MAX20087 chip 1 and 2 towards consumers.

- For each successful I2C communication, CPDD shall update the I2C reading and the corresponding timestamp.
- In case of I2C communication failure, CPDD shall not update the corresponding data or timestamp.

REQ-347241v4 CPDD Interval-based Monitoring periodicity

General comments: Periodicity to monitor faults and register values.
Verification Method: Unit testing, integration testing

The Camera Power Device Driver (CPDD) shall monitor MAX20087 chips at an interval of 100 ms.

REQ-343286v5 Cameras Power Switch Output Current

Purpose: The load current delivered to each camera shall be logged in a Diagnostic Identifier (DID) to provide accurate and actionable data for fault tracing and diagnostics. This information will be available for retrieval to support troubleshooting, fault analysis.

Verification Method: Unit testing, Integration testing

The SW Element Camera Power Diagnostics Driver (CPDD) shall record and store the output current for each channel and make available for consumers.

REQ-343290v8 Cameras Power Switch Output Voltage

Purpose: To support monitoring and diagnostics of the camera power subsystem by providing accurate, timestamped voltage information when needed.

Rationale: Reading the voltage for each camera requires multiple read/write opperations on a shared I2C bus since it requires switching reading current to voltage. Since no diagnostics requires the voltage to be continuously monitored this is an uneccesary load on the bus that should be avoided.

Verification Method: Unit testing, Integration testing

The Camera Power Device Driver (CPDD) shall provide the following power-supply data for each camera channel whenever requested by the Camera Fault Monitor (CFM):

- Output voltage, expressed in millivolts (mV).
- Timestamp indicating when the voltage measurement was recorded.

REQ-344351v5 Cameras Power Switch Channel Status

General comments: Record the fault status of each power supply output for Cameras

Purpose: The purpose of this requirement is to ensure reliable power supply and protect the connected camera modules by continuously monitoring each channel's status. By monitoring the Thermal Shutdown, Over Current, Over Voltage, and Under Voltage conditions.

Verification Method: Unit testing, integration testing

The SW Element Camera Power Diagnostics Driver (CPDD) shall read and monitor the following statuses for each channel of the power switch:

- Thermal Shutdown
- Over Current
- Over Voltage
- Under Voltage

REQ-344353v5 Cameras Power Switch Enable Status

Purpose: The purpose of monitoring and logging the Enable status of each power switch channel is to ensure visibility into the current operational state of the camera power supplies. This data is essential for fault tracing, camera system diagnostics, and verifying the power supply management across individual camera channels.

Verification Method: Unit testing, integration testing

The SW Element Camera Power Diagnostics Driver (CPDD) shall record and store the Enable status of each power switch channel and make available for consumer(s).

REQ-347240v5 Cameras Power Switch Register data

General comments: MAX20087 registers value to be used by consumer(s) for diagnostics.

Purpose: The purpose of this requirement is to ensure that the SW Element Camera Power Device Driver (CPDD) provides access to critical diagnostic registers within the MAX20087 chip. This access enables the detection, monitoring, and reporting of various hardware faults, such as undervoltage, overvoltage, thermal shutdown, ISET faults, and overcurrent/undercurrent conditions.

By exposing the relevant registers and preprocessed data, The SWC CDM can reliably identify and report these faults to the appropriate consumer(s), ensuring proper diagnostic functionality and fault handling for the Power switch MAX 20087

Verification Method: Unit testing, Integration testing

The Camera Power Device Driver (CPDD) shall make all relevant registers and preprocessed information available to detect faults from the MAX20087 chip. This includes, but is not limited to:

- STAT1 and STAT2: For detecting internal faults such as undervoltage, overvoltage, thermal shutdown, and ISET faults.

- CONFIG: For monitoring configuration-related issues that could lead to incorrect operation.
- MASK: To control the fault masking for unused GSML2 ports and prevent unnecessary interrupts.
- ADC[1:4]: To read preprocessed current and voltage information necessary for detecting overcurrent, undercurrent, and voltage out-of-range faults.

The SWE CPDD shall ensure that all relevant fault-related information from these registers is available to the consumer(s) for fault detection and reporting.

REQ-353146v6 CPDD - Camera Safety Switch

General comments: This requirement guarantees that the cameras are protected from running under unsafe circumstances.The shutdown is initiated by a request from the CFM.

Purpose: To have controllability of camera power

Verification Method: Verify that the CPDD turn off the cameras when requested by CFM

The SW Element Camera Power Device Driver (CPDD) shall turn off the power supply to a specific camera upon receiving a request from the Camera Fault Monitor (CFM).

Note: See reference REQ - 345362 Camera Channel Mapping

REQ-445522v3 CPDD - Hardware faults implementation

Purpose: To ensure that the Camera Power Device Driver (CPDD) correctly reports hardware faults from the MAX power switch chip to consumers, while avoiding false fault triggers.

The Software Element Camera Power Device Driver (CPDD) shall monitor the MAX Power chip and propagate the fault information to consumers as per the expressions below for Open Circuit, Short to Battery, and Short to Ground faults.

Open Circuit = {EN ==1 && Iout < Threshold && (STB == 0 && STG == 0)}

Short to Ground = {EN ==1 && (UV ==1 || OC ==1)}

Short to Battery = {EN ==1 && OV ==1}

Note:

- All faults require 3 consecutive fault frames before propagating the fault to consumers.
- The fault will only be set or cleared for powered cameras (EN bit == 1).

Logic flow has been updated in the design guideline REQ-445663

REQ-345362v6 Design Guideline: Camera Channel Mapping

General comments: Refer the document "VCUv2_high_Electrical Schematic_B01_V01", Schematics been uploaded in the below path
https://intranet.volvocars.net/:f:/r/sites/ARTADADASPlatform/Shared%20Documents/Room%20-%20Team%20Exovision/VCUv2_Schematics?csf=1&web=1&e=D9dsq5

This Design Guideline provides an overview of the configuration of the MAX20087 Power Switch Chip in the VCUv2, VCUv2 HI and ADPM, detailing which power switch channels are connected to each camera. The configurations are organized into three sets: Config 1, Config 1.5, and Config 2. Each configuration specifies the MAX chip, power switch channel, output, and the connected camera for precise mapping across the camera system.

Config 1:

VCU Orin Based (VCU V2)	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	Short Range Camera Front
		2	U1_120	OUT2	Short Range Camera Left
		3	U1_120	OUT3	Short Range Camera Right
		4	U1_120	OUT4	Short Range Camera Back
	2	1	U1_121	OUT1	Driver Monitoring Camera D
		2	U1_121	OUT2	Driver Monitoring Camera I
		3	U1_121	OUT3	NC

		4	U1_121	OUT4	Forward Looking Camera Wide

Config 1.5:

VCUv2 HI	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	Short Range Camera Front
		2	U1_120	OUT2	Short Range Camera Left
		3	U1_120	OUT3	Short Range Camera Right
		4	U1_120	OUT4	Short Range Camera Back
	2	1	U1_121	OUT1	Driver Monitoring Camera D
		2	U1_121	OUT2	Driver Monitoring Camera I
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	Forward Looking Camera Wide
	3	1	U1_119	OUT1	Long Range Back Camera Right
		2	U1_119	OUT2	Long Range Back Camera Left
		3	U1_119	OUT3	Long Range Intersection Camera Right
		4	U1_119	OUT4	Long Range Intersection Camera Left

Config 2:

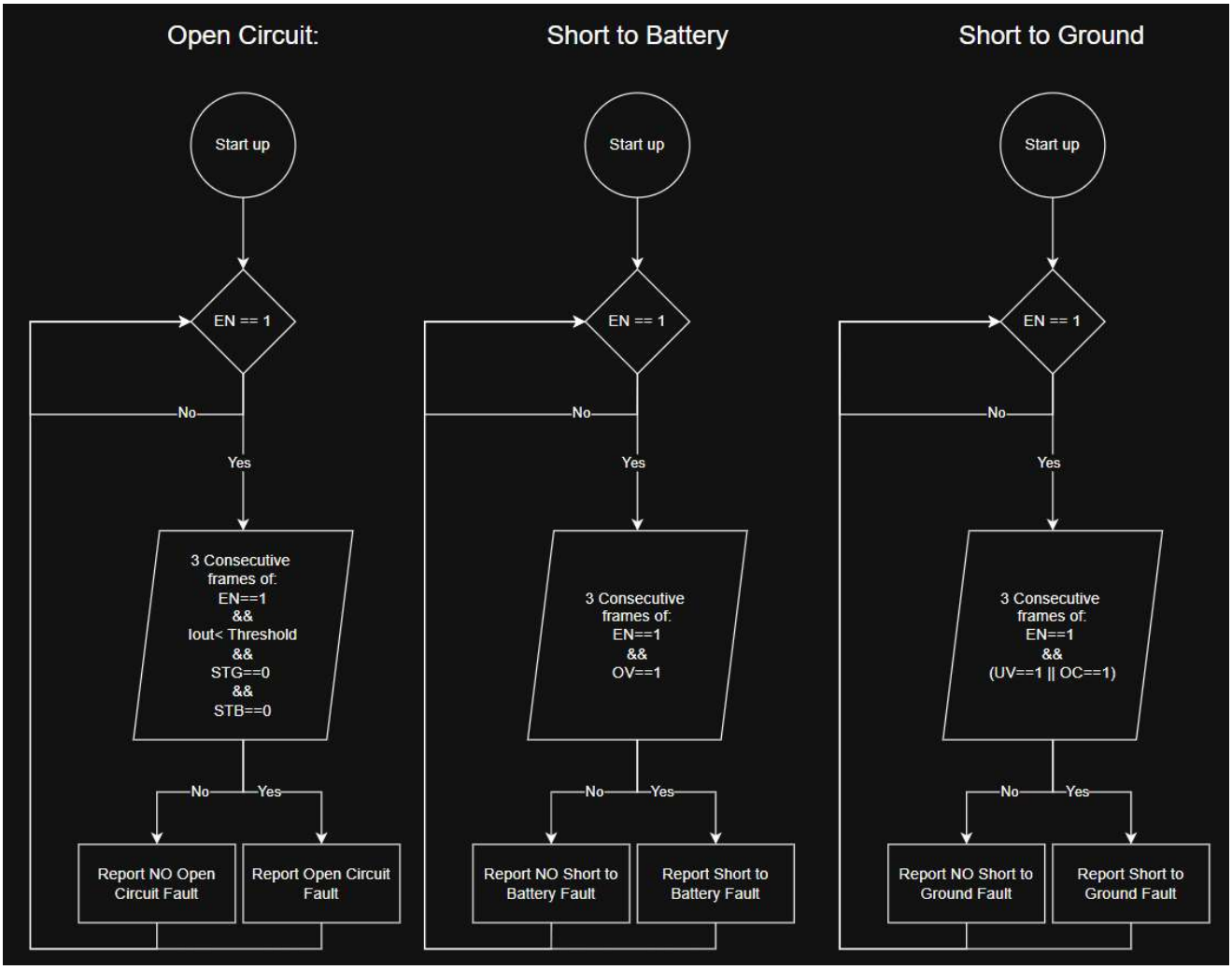
VCU Orin Based (VCU V2)	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	Short Range Camera Front
		2	U1_120	OUT2	Short Range Camera Left
		3	U1_120	OUT3	Short Range Camera Right
		4	U1_120	OUT4	Short Range Camera Back
	2	1	U1_121	OUT1	Driver Monitoring Camera D
		2	U1_121	OUT2	Driver Monitoring Camera I
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	Forward Looking Camera Wide

ADPM	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	NC
		2	U1_120	OUT2	NC
		3	U1_120	OUT3	NC
		4	U1_120	OUT4	NC
	2	1	U1_121	OUT1	Long Range Back Camera Left
		2	U1_121	OUT2	Long Range Back Camera Right
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	NC

NC**: No connection

REQ-445663v3 Design Guideline: CPDD - Logic for Hardware faults

CPDD shall use the below logic flow to define the Hardware faults for MAX 20087 Chip.



Legend:

EN → Enable bit

UV → Under voltage

OC → Over Current

OV → Over Voltage.

Refer to link:

https://confluence.volvocars.biz/spaces/ADVCCZC/pages/937506956/Hardware+Logic+for+CPDD+and+CFM+faults

REQ-363318v2 Design Guideline: Power Switch MAX 20087 - Register Map

Refer the Datasheet of MAX 20087 for further detailed information.

REGISTER	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0	CMD	R/W	POWER-ON RESET
MASK	OVTST	ACCM	TSM	VDDM	VINM	OCM	OVM	UVM	0x00	R/W	0x00
CONFIG	MUX1	MUX0	ENC	CLR	EN4	EN3	EN2	EN1	0x01	R/W	0x1F
ID	—	—	ID[5:4]		R[3:0]				0x02	R	0x00
STAT1	—	—	ISET	ACC	OVIN	UVIN	OVDD	UVDD	0x03	R	0x00
STAT2	TS2	OC2	OV2	UV2	TS1	OC1	OV1	UV1	0x04	R	0x00
	TS4	OC4	OV4	UV4	TS3	OC3	OV3	UV3	0x05	R	0x00
ADC1	D[7:0]								0x06	R	0x00
ADC2	D[7:0]								0x07	R	0x00
ADC3	D[7:0]								0x08	R	0x00
ADC4	D[7:0]								0x09	R	0x00

1.2 PC ECDP - Towards SW Element Camera Fault Monitor (CFM)

REQ-360222v4 CFM - AD&ADAS configuration handling

Purpose: To ensure that camera diagnostics are implemented on the correct camera channels across configurations.

The Software Element Camera Fault Monitor (CFM) in Software Component Camera Diagnostic Monitor (CDM) shall only implement diagnostics for the camera channels populated for the active vehicle configuration.

REQ-396837v3 CFM - Fault maturation process

Purpose: The purpose of these requirements is to define when and how the Camera Fault Monitor (CFM) should mature camera faults and communicate with the Diagnostic Kernel. These requirements ensure reliable camera operation by differentiating between temporary issues that may self-resolve and critical failures requiring immediate system attention, thereby maintaining system integrity while minimizing false alarms.

Rationale: Robust fault detection and maturation mechanisms are necessary because:

1. Camera failures must be detected promptly to maintain system safety and reliability
2. False positives must be minimized to prevent unnecessary system degradation
3. Different fault conditions require different response strategies based on severity and recoverability
4. An incremental counter approach provides resilience against transient faults while responding quickly to persistent issues

The graduated counter approach for Self-Healing and Relinking states allows:

1. Tolerance for intermittent issues that may self-resolve
2. Progressive response to persistent problems
3. Filtering of false positives through counter decay

The immediate maturation for Deactivated state is necessary because:

1. The Deactivated state indicates a critical fault condition determined by the Camera Service
2. No recovery attempt is possible without external intervention
3. Immediate reporting ensures prompt system response to critical camera failures

Fault Maturation for Self-Healing and Relinking States

The Camera Fault Monitor (CFM) shall mature a specific fault ID when both of the following conditions are true:

- The Camera Service status is not any of the following: "Operational", "Deactivated" or "Cooldown"
- The counter for the specific fault ID reaches the threshold value of "x"

"x" shall be defined as below,
x = (faults_per_relinking_attempts * number_of_relinking_attempts * increment_value * desired_fault_percentage)
y = increment value

Counter Management

The Camera Fault Monitor (CFM) shall:

- Process frames every 100 ms
- Increment the counter by "**y**" when:
  - The camera state is Self-Healing or Relinking AND
  - a fault frame is detected
- Decrement the counter by 1 when :
  - The camera state is Self-Healing, Relinking, Operational or Cooldown AND
  - a non-fault frame is detected
- Prevent the counter from exceeding "**x**" (upper bound)
- Prevent the counter from falling below 0 (lower bound)

Actions:

When counter reaches "**x**", CFM shall:

- Mature the specific fault ID
- Send **TestFailed** status for this fault ID to Diagnostic Kernel

Fault Maturation for Deactivated State

The CFM shall mature the specific fault ID when:

- The Camera Service reports a "Deactivated" status for this fault ID

Actions:

When Deactivated state is detected, CFM shall:

- Immediately mature the fault ID without counter validation
- Send **TestFailed** status to Diagnostic Kernel

Fault Maturation for Cooldown State

The CFM shall increment the counter for the over temperature fault ID when either:

- The Camera Service reports a "Cooldown" status OR
- The Camera Service reports an over temperature fault.

The CFM shall mature the specific fault ID when the counter reaches the threshold value.

Notes:

- The CFM shall track each fault ID independently through the maturation process
- Actions taken for one fault ID shall not affect the maturation state of other fault IDs
- Camera state priority: Deactivated > Cool-down > Relinking > Self-healing

==========
==========

Pseudo Code Logic for reference:

```
if (frame_has_fault == true) {
    // Increment allowed only in non-operational & non-deactivated states
    if (camera_state == SELFHEALING || camera_state == RELINKING || camera_state == COOLDOWN ) {
        counter = min(counter + y, x); // Increment but saturate at x
    }
    //Cooldown/Overtemp handling
    else if (camera_state == COOLDOWN || overTempFaultReported()) {
        counter = min(counter + y, x); // Increment but saturate at x
    }
} else {
    if (camera_state == SELFHEALING ||
        camera_state == RELINKING ||
        camera_state == OPERATIONAL ||
        camera_state == COOLDOWN ) {
            // Decrement but not below 0
```

```
        counter = max(counter - 1, 0);
    }
}

// Fault maturation
if (camera_state != OPERATIONAL && camera_state != DEACTIVATED) {
    if (counter == x) {
        send_test_failed_to_diagnostic_kernel(fault_id);
    }
}

// Deactivated override
if (camera_state == DEACTIVATED) {
    send_test_failed_to_diagnostic_kernel(fault_id);
}
```

REQ-396838v2 CFM - Fault de-maturation process

Purpose: To enable automatic recovery from temporary camera faults by allowing the system to safely demature such faults, thereby sustaining continuous camera operation and minimizing unnecessary downtime.

Rationale: The graduated counter dematuration approach ensures:

1. Confirmation of stable camera operation before clearing faults
2. Prevention of fault oscillation in borderline conditions

The immediate dematuration for service intervention is necessary because:

1. It enables service technicians to verify repairs
2. It allows for system reset after component replacement
3. It provides a controlled method to clear persistent faults

The Camera Fault Monitor (CFM) shall demature a fault when either:

- The Camera Service status returns to "Operational" state OR
- The counter for the specific fault ID reaches the threshold value of 0

Counter Management

The CFM shall both:

- Decrement the counter by 1 when no fault is detected in a frame AND
- Prevent the counter from falling below 0 (lower limit)

Actions:

When the counter reaches 0, the CFM shall both:

- Demature the specific fault ID AND
- Send a "**TestPassed**" status for this fault ID to the Diagnostic Kernel.

REQ-341430v9 CFM: Internal Fault on the Deserializer

General comments: Refer to the strategy F from REQ-355276 Design Guidline 'Camera Diagnostic strategy'

Purpose: To ensure that the Camera Fault Monitor (CFM)propagates Deserializer faults detected by the Software Component Camera Service to relevant consumers (such as Core System Platform Diagnostics/SWC HP Platform Software).

Verification Method: Unit testing, Integration testing

- The Camera Fault Monitor shall propagate internal faults in deserializers when notified by the Software Component (SWC) Camera Service to consumers.
- The Camera Fault Monitor shall raise an internal fault flag when the SWC Camera Service detects an internal fault in deserializers and pass this information via an error interface to consumers.
- The Camera Fault Monitor shall not mature the internal faults in deserializers into a DTC and shall not handle

any DTC-related state transitions.

- Fault information related to the deserializer shall be reported on an event basis (as per REQ-363562)

Note:
Detection criteria of internal fault in MAX96712 Deserializer:

- *No communication with Deserializer while communication with corresponding Power Switch Chip is functioning (see error_api from CPDD REQ-339734).*
- *The Error API in Camera Service reports an internal fault in Deserializer*

REQ-341431v6 CFM - Fault during Initialization

General comments: Refer to the strategy A from REQ-355276 Design Guidline 'Camera Diagnostic strategy'

Handshake - other party status : Closed

Precondition: No fault from Camera Power Switch is reported.

Purpose: Set a DTC if a SW fault was identified during the init phase by Camera Service.

Verification Method: Unit testing, Integration testing

Fault Maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault maturation logic as defined in REQ-396837 during the initialization.
When fault maturation criteria are met, the CFM shall:

- Set the "**TestFailed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to set the Diagnostic Trouble Code (DTC) **xxxx93** (where xxxx is defined in REQ-341424)
- Request DK to set the local snapshot information of the DTC as per the fault information as per the design guideline REQ-427637.

Fault De-maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault de-maturation logic as defined in REQ-396838 during the initialization phase.
When fault de-maturation criteria are met, the CFM shall:

- Set the "**TestPassed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to clear the Diagnostic Trouble Code (DTC) **xxxx93**

Reference:

- The CFM shall apply the fault maturation and de-maturation logic continuously throughout the initialization phase.
- The CFM shall update the Diagnostic Kernel status within the timing constraints specified in REQ-396837 and REQ-396838.

REQ-341432v7 CFM - Un recoverable Fault during Runtime

General comments: Refer to the strategy B from REQ-355276 Design Guidline 'Camera Diagnostic strategy'

Handshake - other party status : Closed

Precondition: 1. No fault from Camera Power Switch is reported.

Verification Method: Unit testing, Integration testing

Fault Maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault maturation logic as defined in REQ-396837 during the Runtime (non recoverable run time faults).

When fault maturation criteria are met, the CFM shall:

- Set the "**TestFailed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to set the Diagnostic Trouble Code (DTC) **xxxx96** (where xxxx is defined in REQ-341424)
- Request DK to set the local snapshot information of the DTC as per the fault information as per the design guideline REQ-427637.

Fault De-maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault de-maturation logic as defined in REQ-396838 during the runtime phase.
When fault de-maturation criteria are met, the CFM shall:

- Set the "**TestPassed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to clear the Diagnostic Trouble Code (DTC) **xxxx96**

Reference:

- The CFM shall apply the fault maturation and de-maturation logic continuously throughout the initialization phase.
- The CFM shall update the Diagnostic Kernel status within the timing constraints specified in REQ-396837 and REQ-396838.

REQ-341433v7 CFM- Recoverable Fault during Run time

General comments: Refer to the strategy C from REQ-355276 Design Guidline 'Camera Diagnostic strategy'
Handshake - other party status : Closed
Verification Method: Unit testing, Integration testing

Fault Maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault maturation logic as defined in REQ-396837, for recoverable run time faults been processed by Camera Service,
When fault maturation criteria are met, the CFM shall:

- Set the "**TestFailed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to set the Diagnostic Trouble Code (DTC) **xxxx92** (where xxxx is defined in REQ-341424)
- Request DK to set the local snapshot information of the DTC as per the fault information as per the design guideline REQ-427637.

Fault De-maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault de-maturation logic as defined in REQ-396838,
When fault de-maturation criteria are met, the CFM shall:

- Set the "**TestPassed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to clear the Diagnostic Trouble Code (DTC) **xxxx92**

Reference:

- The CFM shall apply the fault maturation and de-maturation logic continuously throughout the initialization phase.
- The CFM shall update the Diagnostic Kernel status within the timing constraints specified in REQ-396837 and REQ-396838.

REQ-341434v8 CFM - Fault on Camera Over Temperature

General comments: Refer to the strategy D from REQ-355276 Design Guidline 'Camera Diagnostic strategy'
Handshake - other party status : Closed
Precondition: The Precondition serves as additional contextual information that enhances the overall meaning of the Requirement (as detailed in the Description section). It is not intended to be a

separate requirement but rather acts as a guideline for the CFM, indicating the diagnostic strategy to be employed.

- No active fault is reported from the Camera Power Switch
- No active Deserializer fault is reported

Verification Method: Unit testing, Integartion testing

Fault Maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault maturation logic(cooldown state) as defined in REQ-396837
When fault maturation criteria are met, the CFM shall:

- Set the "**TestFailed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to set the Diagnostic Trouble Code (DTC) **xxxx98** (where xxxx is defined in REQ-341424)

Fault De-maturation Logic:

The Camera Fault Monitor (CFM) shall apply the fault de-maturation logic as defined in REQ-396838,
When fault de-maturation criteria are met, the CFM shall:

- Set the "**TestPassed**" status in the Diagnostic Kernel
- Instruct the Diagnostic Kernel to clear the Diagnostic Trouble Code (DTC) **xxxx98**

Reference:

- The CFM shall apply the fault maturation and de-maturation logic continuously throughout the initialization phase.
- The CFM shall update the Diagnostic Kernel status within the timing constraints specified in REQ-396837 and REQ-396838.

REQ-339740v7 CFM - Freeze Frame data

Purpose: To ensure that critical parameters and contextual data are captured as a freeze frame during a camera fault, providing detailed insight for fault tracing and diagnostics. This supports understanding the system state at the time of failure and enables effective problem resolution.

When the Camera Fault Monitor (CFM) reports a DTC testFailed event due to a fault reported by Camera Service, the Camera Fault Monitor (CFM) shall:

1. Create a freeze frame (snapshot) associated with the reported DTC, including the fault category recieved from Camera Service.
2. Store the fault category in a Data Identifier (DID).

The DID used for the Freeze Frame Data shall be according to REQ-479173.

REQ-347794v6 CFM - Fault Over voltage

General comments: Refer to the strategy E from REQ-355276 Design Guidline 'Camera Diagnostic strategy'

Precondition: The Precondition serves as additional contextual information that enhances the overall meaning of the Requirement (as detailed in the Description section). It is not intended to be a separate requirement but rather acts as a guideline for the CFM, indicating the diagnostic strategy to be employed.

1. No fault from Camera Power Switch is reported.
2. No deserialyzer fault is reported.

Verification Method: Unit testing, Integartion testing

The Camera Fault Monitor (CFM) shall send **TestFailed** and request the Diagnostic Kernel (DK) to set the Diagnostic Trouble Code (DTC) xxxx9A (where xxxx is represented in REQ-341424) immediately upon receiving the first fault frame from the Camera Power Device Driver (CPDD).

- Detection Criteria:
  - testFailed: The camera Over Voltage flag is informed by CPDD.
  - testPassed: No faults are reported from CPDD.

REQ-349578v6 CFM - ISET fault monitor

General comments: ISET is responsible for setting the current limit for the power switch MAX20087 chip. It regulates the amount of current that flows through the connected camera components.
An ISET fault occurs when the resistance setting for the current regulation is out of range, which can lead to uncontrolled current flowing through the system.

Purpose: To prevent CFM from incorrectly flagging other components as faulty (e.g., camera, sensors) due to incorrect current regulation.

The Software Element Camera Fault Monitor (CFM) shall detect and respond to an ISET fault reported by the Software Component Camera Power Driver Diagnostics (CPDD).

- The CFM shall immediately flag an internal fault.
- All DTC tests shall be temporarily interrupted while the ISET fault is active.
- The CFM shall not trigger any DTCs during the ISET fault condition.
- Upon detection of an ISET fault in the Power Switch Chip MAX20087 the CFM shall immediately propagate the fault information to consumers according to **REQ-363562** *'CFM - Consumers of Fault propagation'*

Notes:
Always refer to the latest versions of REQ-363562

REQ-339593v8 CFM - Detection of fault for short to ground

General comments: Purpose
populate the fault information about short to ground fault for channel 1 to 4 of MAX chip 1 and 2.

Precondition: The Precondition serves as additional contextual information that enhances the overall meaning of the Requirement (as detailed in the Description section). It is not intended to be a separate requirement but rather acts as a guideline for the CFM, indicating the diagnostic strategy to be employed.

1. Data provider(s) should provide short to ground information.
2. No ISET fault is identified from Camera Power Switch.
3. No Thermal-shutdown fault is identified from Camera Power Switch.

Verification Method: Unit testing, Integration testing

The Camera Fault Monitor (CFM) shall send **TestFailed** and request the Diagnostic Kernel (DK) to set the Diagnostic Trouble Code (DTC) xxxx11 (where xxxx is represented in REQ-341424) immediately upon receiving the first fault frame from the Camera Power Device Driver (CPDD).

- Detection Criteria:
  - testFailed: The CFM identifies a Short to Ground event in the specific channel from CPDD.
  - testPassed: No faults are reported from CPDD.

REQ-339589v7 CFM - Detection of fault for short to battery

General comments: Purpose
populate the fault information about short to battery fault for channel 1 to 4 of MAX chip 1 and 2

Precondition:
1. Data provider(s) should provide short to battery information.
2. No ISET fault is identified from Camera Power Switch.
3. No Thermal-shutdown fault is identified from Camera Power Switch.

Verification Method: Unit testing, Integration testing

The Camera Fault Monitor (CFM) shall send **TestFailed** and request the Diagnostic Kernel (DK) to set the Diagnostic Trouble Code (DTC) xxxx12 (where xxxx is represented in REQ-341424) immediately upon receiving the first fault frame from the Camera Power Device Driver (CPDD).

- Detection Criteria:
  - testFailed: The CFM identifies a Short to Battery event in the specific channel from CPDD.

- testPassed: No faults are reported from CPDD.

REQ-339553v6 CFM - Detection of fault for open circuit

General comments: Purpose
Populate the fault information about open circuit fault for channel 1 to 4 of MAX Chip 1 and 2

Precondition: Preconditions
1. Data Provider(s) should provide open circuit information.
2. No ISET fault is identified from Camera Power Switch.
3. No Thermal-shutdown fault is identified from Camera Power Switch.

Verification Method: Unit testing, Integration testing

The Camera Fault Monitor (CFM) shall send **TestFailed** and request the Diagnostic Kernel (DK) to set the Diagnostic Trouble Code (DTC) xxxx13 (where xxxx is represented in REQ-341424) immediately upon receiving the first fault frame from the Camera Power Device Driver (CPDD).

- Detection Criteria:
  - testFailed: The CFM identifies an Open Circuit event in the specific channel from CPDD.
  - testPassed: No faults are reported from CPDD.

REQ-339722v8 CFM - Detection of fault for Thermal shutdown

General comments: Set DTC for Thermal shutdown for channel 1 to 4 of MAX chip 1 and 2

Precondition: No I2C communication fault is identified from Camera Power Switch
No ISET fault is identified from Camera Power Switch.

Purpose: The purpose of this requirement is to ensure that the Camera Fault Monitor (CFM) detects thermal shutdown events in the MAX20087 power chip and raises an internal fault flag for consumers. This fault flag will be propagated via an error interface (e.g., Core System Platform Diagnostics/SWC HP Platform Software) to enable further diagnostics and recovery actions. The CFM shall not mature the fault into a DTC but provide relevant fault information to consumers via the error interface

Verification Method: Unit testing, Integration testing

Camera Fault Monitor (CFM) shall detect Thermal shutdown faults in the MAX20087 power chip and propagate fault information through an error API to the relevant consumers.

- The CFM shall raise an internal fault flag when a Thermal shutdown event is detected.
- The CFM shall not mature the Thermal shutdown fault into a DTC and shall not perform any DTC state transitions.
- Fault information shall be provided to consumers through an error interface.
- Upon detection of a Thermal Shurdown the CFM shall immediately propagate the fault information to consumers according to **REQ-363562** *'CFM - Consumers of Fault propagation'*

Note:
Always refer to the latest version of **REQ-363562** *'CFM - Consumers of Fault propagation'*

REQ-349505v6 CFM - Fault information propagation

General comments: The term "relevant consumers" refers to all software components in HPx execution environment that require access to fault status information for effective fault handling and diagnostics. Specifically, this includes, but is not limited to, the Domain Controller in Product Module AAE, the Health Monitor, and potentially the Operational Manager in Product Module ACA. These components are involved in decision-making processes related to AD&ADAS functional safety, reliability, and operational integrity, thus requiring accurate and timely fault information from the Camera Fault Monitor (CFM).

The term 'Confirmed faults (DTC confirmed, service/workshop required)' are those that cannot be resolved without specialized intervention, often requiring workshop repairs such as hardware replacement or system recalibration.

Purpose: To ensure that critical camera-related faults are effectively communicated to the relevant consumers, enabling timely corrective actions, proper fault diagnostics, and system recovery.

Rationale: By ensuring only matured and confirmed camera related faults are propagated, the CFM prevents unnecessary or false alarms, reducing system complexity and the risk of

misinterpreting transient faults. Providing these DTCs to relevant consumers ensures that camera failures are properly logged for diagnostics while enabling necessary operational measures to maintain overall AD&ADAS system integrity.

The SW Element Camera Fault Monitor (CFM) shall monitor the fault status and inform the service required message to the consumers.

When a confirmed DTC been set from Diagnostics Kernel, CFM shall consider the DTC confirmed status and inform the consumers about the service required message for the appropriate confirmed DTC's.

Confirmed fault with Service/workshop status can be categorised as below:

- Confirmed faults (DTC confirmed, service/workshop required)
    - When a permanent and unrecoverable fault is detected, which may be caused by a camera hardware failure, software issue, or problems with associated camera harnesses.
    - Such faults indicate that recovery within normal system operation is not possible, and service intervention (e.g., workshop repair or hardware replacement) is required.
    - The CFM shall propagate the corresponding DTC to the relevant consumers. The fault is confirmed, and no further recovery action can be taken in the system.

- CFM shall provide a timestamp and fault context for each propagated fault, ensuring traceability and detailed analysis.

Note:
For further details on timestamp and fault context, refer to **REQ-339740** *'CFM - Freeze Frame data'* (always use the latest version)
For further details on consumers, refer to **REQ-363562** *'CFM - Consumers of Fault propagation'* (always use the latest version)

REQ-353144v8 CFM - Fault inputs towards CPDD for camera power cut-off

Purpose: To ensure that the Camera Fault Monitor (CFM) can trigger power control actions through the Camera Power Device Driver (CPDD) to protect hardware components.

Rationale: The cameras should only be powered off to revent damage to hardware.
The first condition protects the power switch when there is a fault on the output. The second condition protects the camera when it exceeds the safe opperating temperature and corrective action from camera service fails.

The SW Element CFM (Camera Fault Monitor) shall receive fault active information and camera status per camera from Camera Service application (refer to **REQ-389630**).

CFM shall request CPDD to turn off power to the affected camera(s) when either of the following conditions are met:

Condition 1:

Both:

- The Camera Service operational state for the respective camera is Deactivated AND
- Short to battery OR short to ground fault is reported by the Camera Power Device Driver (CPDD) for the respective camera

Condition 2:

Both:

- kDisablelink is received from Camera Service AND
- The Camera Service reports an over temperature fault OR The Camera Service operational state for the respective camera is Cooldown

Note:
Always refer to the latest version of **REQ-389630** *'Camera Service - Camera status reporting for Fault monitoring'*

REQ-363383v4 CFM - Detection of fault for Undervoltage/Overvoltage Lockout

Purpose: The purpose of this requirement is to ensure that the Camera Fault Monitor (CFM) propagates both Overvoltage and Undervoltage Lockout faults detected by the Camera Power Device Driver (CPDD) via the appropriate registers (e.g., STAT1) to the relevant consumers (such as Core System Platform Diagnostics or SWC HP Platform Software). The CFM will not mature these faults into DTCs but will flag them as internal faults and propagate the fault information through the error interface.

- The CFM shall propagate both Overvoltage and Undervoltage Lockout faults when notified by the Camera Power Device Driver (CPDD)
- The CFM shall raise an internal fault flag when the CPDD detects an Overvoltage Lockout fault or Undervoltage Lockout fault and propagate this fault information via the error API to consumers.
- The CFM shall not mature the Overvoltage or Undervoltage Lockout faults into DTCs and shall not manage any DTC-related state transitions.
- If an Overvoltage or Undervoltage Lockout fault is identified by the CPDD, the CFM shall ignore other faults from the SWC Camera Service while these faults are active.
- Fault information related to both **Overvoltage/Undervoltage** Lockout faults shall be reported to consumers according to **REQ-363562 '***CFM - Consumers of Fault propagation'*

Note:
Always refer to the latest version of **REQ-363562 '***CFM - Consumers of Fault propagation'*

REQ-363562v6 CFM - Consumers of Fault propagation

Purpose: The purpose of this requirement is to ensure immediate detection and propagation of internal faults in the Power Switch MAX20087 and Deserializer by the Camera Fault Monitor (CFM). Fault information is sent to two key consumers:
- CSP Core System Diagnostic (HP Platform Software) for managing system diagnostics.
- Driver Monitoring Vision Processing for handling camera-related faults affecting driver monitoring.

Faults related to the DMCx channels to be propagated to Driver Monitoring Vision Processing

The SW Element Camera Fault Monitor (CFM) shall detect any internal faults related to:
- Power Switch MAX20087
- Deserializer

Consumers of the CFM fault information are:
- CSP Core System Diagnostic - SWC *HP Platform Software* and
- SWC *Driver Monitoring Vision Processing*

Upon detection of any internal fault related to the Power Switch Chip MAX20087 or the Deserializer, the CFM shall immediately propagate the fault information to CSP Core System Diagnostic - SWC *HP Platform Software* AND SWC *Driver Monitoring Vision Processing.*

Upon detection of any fault related to the channels connected to the DMCx cameras, the CFM shall propagate the following fault information to SWC *Driver Monitoring Vision Processing*:

- Open circuit on channels connected to the DMCx cameras.
- Short to battery on channels connected to the DMCx cameras.
- Short to ground on channels connected to the DMCx cameras.
- Supply voltage faults on channels connected to the DMCx cameras.
- Supply current faults on channels connected to the DMCx cameras.

Note: DMC refers for A camera that monitors the driver

REQ-293300v2 Enabling support for Hostlog and ZVS with DCF for Exterior Camera Perception data

Purpose: To support system and software verification for Exterior Camera Perception by enabling detailed analysis of logs from core computers. This will facilitate the diagnosis of faults, improve system reliability, and ensure traceability throughout the development process.

Rationale:

Detailed logging of internal variables and states is essential for diagnosing system behaviors, especially in complex perception systems like Exterior Camera Perception. By leveraging the Data Collection Framework (DCF), development and verification teams can collect valuable insights into the exterior camera system's performance, ensuring early detection of faults and enhancing overall system robustness.

Verification: Verify that internal variables and states can be collected correctly in a test setup that includes a core computer and a logger. Testing in component HIL rig for Active Safety domain. Results to be reported into the Continuous Integration (CI) tool chain as part of System/ART level release verification.

Requirement Description:

- The SW Element Camera Fault Monitor (CFM) shall be responsible for implementing the logging functionality to collect internal variables and states provided by Camera Service application.
- This logging shall be used to support system and software development and verification activities and shall utilize the Data Collection Framework (DCF) to ensure compatibility and standardization across development environments.
- The logging data must facilitate detailed analysis to support design verification, fault identification, and system improvement activities.

REQ-374665v6 CFM - DID Cameras Power Switch Output Voltage

Purpose: To enable consumers to retrieve accurate, time-correlated camera supply voltage information for monitoring and fault analysis.

The Camera Fault Monitor (CFM) shall provide the following information for each camera to the requesting consumers upon the corresponding Data Identifier (DID) read request:

- Camera output voltage, expressed in millivolts (mV).
- Timestamp of when the voltage measurement was taken.

The DID used for the Camera Power Switch Output Voltage shall be according to REQ-479173

Note: Refer *REQ-343290 Cameras Power Switch Output Voltage*

REQ-374672v5 CFM - DID Camera Power Switch Output Current

General comments: To enable consumers to retrieve accurate, time-correlated camera supply current information for monitoring and fault analysis.

The Camera Fault Monitor (CFM) shall provide the following information for each camera to the requesting consumers upon the corresponding Data Identifier (DID) read request:

- Camera output current, expressed in milliamperes (mA).
- Timestamp of when the current measurement was taken.

The DID used for the Camera Power Switch Output Current shall be according to REQ-479173

Note: Refer *REQ-343286 Cameras Power Switch Output Current*

REQ-374679v4 CFM - DID Cameras Power Switch Output Status

General comments: This information supports logging of Power Switch output status flags, including Thermal Shutdown, Over-Current, Over-Voltage, and Under-Voltage conditions, using a Diagnostic Identifier (DID).

Purpose: To enable consumers to retrieve camera power switch channel status information in support of fault tracing and diagnostics.

The Camera Fault Monitor (CFM) shall provide the following information for each camera to the requesting consumers upon the corresponding Data Identifier (DID) read request:

- Camera power switch output status.
- Timestamp of when the status was read.

The DID used for the Cameras Power Switch Status shall be according to REQ-479173

Note: Refer *REQ-344351 Cameras Power Switch Channel Status*

REQ-374680v4 CFM - DID Cameras Power Switch Enable Status

Purpose: To enable consumers to retrieve camera power switch channel enable status information in support of fault tracing and diagnostics.

The Camera Fault Monitor (CFM) shall provide the following information for each camera to the requesting consumers upon the corresponding Data Identifier (DID) read request:

- Camera power switch channel enable status.
- Timestamp of when the status was read.

The DID used for the Cameras Power Switch Enable Status shall be according to REQ-479173

Note: Refer *REQ-344353 Cameras Power Switch Enable Status*

REQ-355930v4 CFM - Camera Power Control

General comments: Record and let possible to temporarily control the enable/disable status of each power supply output for Cameras

Purpose: The purpose of the requirement is Power Switches control of the channels during service requests.

Verification Method: Unit testing, integration testing

The SW Element CFM shall request CPDD to turn off power to the specified camera channel(s).if the control routine (2F service) requests a power-off through Diagnostic Identifier (DID) D931,

REQ-375331v1 CFM - Requesting DTC towards Diagnostic Kernel

General comments: Interface "Fault Management Library HPA 2.1"
- Interface ID: IF-18898
- Link: url:swap://SystemWeaver:3000/x0400000002F525C2
- Provided by SWC Diagnostic Kernel Application A 2.1, Specific Module Diagnostics 2.1, part of Product Module 'Diagnostics and Fault Management Services'

Purpose: The purpose of this requirement is to ensure the Camera Fault Monitor (CFM) communicates fault states effectively with the Diagnostic Kernel (DK), enabling accurate logging of DTCs in accordance with detection criteria defined in SSDS (CarCom database file extract).

The CFM shall utilize the Diagnostic Kernel (DK) via the interface 'Fault Management Library HPA 2.1' to request the states setting of the following DTCs:

- Fault during Initialization (INIT) Phase:
  - DTC Code: xxxx93
  - Description: Fault detected during the initialization phase.
  - Reference Requirement: REQ-341431

- Fault during Runtime Phase:
  - DTC Code: xxxx96
  - Description: Fault detected during the runtime phase.
  - Reference Requirement: REQ-341432

- Fault - Frame Degradation:
  - DTC Code: xxxx92
  - Description: Frame degradation fault detected during image processing.

    - Reference Requirement: REQ-341433

- Fault - Camera Over-Temperature:
    - DTC Code: xxxx98
    - Description: Over-temperature fault detected in the camera.
    - Reference Requirement: REQ-341434

- Fault - Over Voltage:
    - DTC Code: xxxx9A
    - Description: Over-voltage detected in the power supply to the camera.
    - Reference Requirement: REQ-347794

- Fault - Short to Ground:
    - DTC Code: xxxx11
    - Description: Short to ground detected in the power line.
    - Reference Requirement: REQ-339593

- Fault - Short to Battery:
    - DTC Code: xxxx12
    - Description: Short to battery detected in the power line.
    - Reference Requirement: REQ-339589

- Fault - Open Circuit in Harness:
    - DTC Code: xxxx13
    - Description: Open circuit fault detected in the camera harness.
    - Reference Requirement: REQ-339553

Note: Refer *REQ-341424 Camera Diagnostics IDs* for detailed DTC IDs

REQ-380318v3 CFM - DID Camera Part Number

Purpose: To enable consumers to retrieve camera part number information

The Camera Fault Monitor (CFM) shall provide the following information for each camera to the requesting consumers upon the corresponding Data Identifier (DID) read request:

- Camera part numbers from Camera Service application.
- Timestamp of when the part number was read.

The DID used for the Camera Part Numbers shall be according to REQ-479173

REQ-380320v3 CFM - DID Camera Serial Number

Purpose: To enable consumers to retrieve camera serial number information

The Camera Fault Monitor (CFM) shall provide the following information for each camera to the requesting consumers upon the corresponding Data Identifier (DID) read request:

- Camera serial numbers from Camera Service application.
- Timestamp of when the serial number was read.

The DID used for the Camera Serial Numbers shall be according to REQ-479173

REQ-479173v1 Design Guideline: CFM - DID/RID ID Mapping

General comments: This mapping defines the Data Identifier (DID) values used across different hardware platforms and applications. The differences in DID assignments support different configurations.

Purpose: To define the ID's used by the different Data Identifiers (DIDs) in different applications.

The mapping of DID/RID ID shall be as follows:

Description	Cluster A on HPA	Cluster B on HPA	Cluster B on HPB
Cameras Power Switch Enable Status	0xD931	0x2151	0xD931
Cameras Power Switch Output Voltage	0xD932	0x2152	0xD932
Camera Power Switch Output Current	0xD933	0x2153	0xD933
Cameras Power Switch Channel Status	0xD934	0x2154	0xD934
Freeze Frame Data	0xD0FD	0x2150	0xD0FD
Camera Part Number	0xF13B	0xD996	0xD996
Camera Serial Number	0xF13F	0xD997	0xD997

Note:
Cluster A consists of: FLCW and SRCx
Cluster B consists of: LRIx and LRBx

REQ-360233v3 Design Guideline: AD&ADAS configuration parameters consumption

Configuration Parameter

Name: Assist Drive Redundancy
Code: 0HBA

Name	Description	Config
Level 2-1	Redundancy for supervised self-driving, open for additional redundancy configurations within the level.	1
Level 2-2	Assisted Drive Redundancy Level for Config 1.5	1.5
Level 4-1	Redundancy for unsupervised self-driving, open for additional redundancy configurations within the level.	2

REQ-341424v2 Design Guideline: Camera Diagnostics IDs

Verification Method: Unit testing, Integration testing

The DTC ID for each of the cameras is:

Camera	Production DTC ID
FLCW	D4D5
SRCF	D50A
SRCL	D50B
SRCR	D50C
SRCB	D509
LRBL	D4EE
LRBR	D4EF
LRIL	D606
LRIR	D607

Note: Ensure this information, the YAML configuration and the CARCOM database is consistent.

REQ-427637v3 Design Guideline: Camera Fault categorization from Camera Service

Handshake - other party status : Closed

Below table defines the fault category for the appropriate DTC's

Fault Category	Fault Type	DTC
1. Initialization and Startup Fault	**Initialization & Start-up faults**	**xxxx93**
kSiplCameraInit kSiplCameraStart kInitTimeout kModuleInterfaceCreationFailure kSensorPowerUpSmErrCheckFailure		

kReadEeprom		
2. GPIO Test Fault	**Initialization & Start-up faults**	**xxxx93**
kDeviceBlockNotifErrorGpioTestFailure kDeviceBlockNotifErrorGpioTestSensorFaultInjectionFailure kDeviceBlockNotifErrorGpioTestSerializerFaultInjectionFailure		
3. Serializer and Deserializer Fault	**Run-time unrecoverable faults**	**xxxx96**
kDeviceBlockNotifErrorDeserializerFailure kDeviceBlockNotifErrorSerializerFailure kDeviceBlockNotifErrorGetDeserializerErrorInfoFailure kDisableLink kEnableLink		
4. Sensor and Module Fault	**Run-time unrecoverable faults**	**xxxx96**
kDeviceBlockNotifErrorSensorFailure kDeviceBlockNotifErrorGetModuleErrorInfoFailure		
5. General Device Block Fault	**Run-time unrecoverable faults**	**xxxx96**
kDeviceBlockNotifErrorInternalFailure kDeviceBlockNotifErrorUnknown kDeviceBlockNotifErrorIntrLocalizationFailure kDeviceBlockNotifErrorIntrLocalizationTimeout		
6. Image Processing Pipeline Fault	**Run-time recoverable faults**	**xxxx92**
kPipelineNotifErrorIcpBadInputStream kPipelineNotifErrorIcpCaptureFailure kPipelineNotifErrorIcpEmbDataParseFailure kPipelineNotifErrorIspProcessingFailure kPipelineNotifErrorAcpProcessingFailure kPipelineNotifErrorCdiSetSensorCtrlFailure kIcpGetFrame kIspGetFrame kPipelineNotifWarningCaptureTimeout		
7. Image and Data Integrity Fault	**Run-time recoverable faults**	**xxxx92**
kCameraFrozenImage kParseImageMetadata kEmbeddedDataErrorFlag		
8. Environmental Fault	**Environmental Faults**	**xxxx98**
kOverTemperatureFault kUnderTemperatureFault		

9. System Fault	**System & unknown faults**	**xxxx9A**
kWatchdog kUnknownError kVoltageError kImageTimestampDeviationExceeded kImageTimestampIntegrityFalse		
10. No Fault (Default)		
Any other/unrecognized reason not explicitly listed above.		

xxxx refer to as below:

FLCW	D4D5
SRCB	D509
SRCF	D50A
SRCL	D50B
SRCR	D50C
LRBL	D4EE
LRBR	D4EF
LRIL	D606
LRIR	D607

REQ-345362v6 Design Guideline: Camera Channel Mapping

General comments: Refer the document "VCUv2_high_Electrical Schematic_B01_V01", Schematics been uploaded in the below path
https://intranet.volvocars.net/:f:/r/sites/ARTADADASPlatform/Shared%20Documents/Room%20-%20Team%20Exovision/VCUv2_Schematics?csf=1&web=1&e=D9dsq5

This Design Guideline provides an overview of the configuration of the MAX20087 Power Switch Chip in the VCUv2, VCUv2 HI and ADPM, detailing which power switch channels are connected to each camera. The configurations are organized into three sets: Config 1, Config 1.5, and Config 2. Each configuration specifies the MAX chip, power switch channel, output, and the connected camera for precise mapping across the camera system.

Config 1:

VCU Orin Based (VCU V2)	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	Short Range Camera Front
		2	U1_120	OUT2	Short Range Camera Left
		3	U1_120	OUT3	Short Range Camera Right
		4	U1_120	OUT4	Short Range Camera Back
	2	1	U1_121	OUT1	Driver Monitoring Camera D
		2	U1_121	OUT2	Driver Monitoring Camera I
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	Forward Looking Camera Wide

Config 1.5:

VCUv2 HI	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	Short Range Camera Front
		2	U1_120	OUT2	Short Range Camera Left
		3	U1_120	OUT3	Short Range Camera Right
		4	U1_120	OUT4	Short Range Camera Back

	2	1	U1_121	OUT1	Driver Monitoring Camera D
		2	U1_121	OUT2	Driver Monitoring Camera I
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	Forward Looking Camera Wide
	3	1	U1_119	OUT1	Long Range Back Camera Right
		2	U1_119	OUT2	Long Range Back Camera Left
		3	U1_119	OUT3	Long Range Intersection Camera Right
		4	U1_119	OUT4	Long Range Intersection Camera Left

Config 2:

VCU Orin Based (VCU V2)	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	Short Range Camera Front
		2	U1_120	OUT2	Short Range Camera Left
		3	U1_120	OUT3	Short Range Camera Right
		4	U1_120	OUT4	Short Range Camera Back
	2	1	U1_121	OUT1	Driver Monitoring Camera D
		2	U1_121	OUT2	Driver Monitoring Camera I
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	Forward Looking Camera Wide

ADPM	MAX Chip	Channel	Power Switch	Output	Connected camera
	1	1	U1_120	OUT1	NC
		2	U1_120	OUT2	NC
		3	U1_120	OUT3	NC
		4	U1_120	OUT4	NC
	2	1	U1_121	OUT1	Long Range Back Camera Left
		2	U1_121	OUT2	Long Range Back Camera Right
		3	U1_121	OUT3	NC
		4	U1_121	OUT4	NC

NC**: No connection

REQ-363249v1 Design Guideline: CFM - Fault Threshold in Power Switch Max 20087

Purpose: To ensure consistent and accurate fault detection, CFM must implement thresholds for various fault types using the values specified in the MAX20087 datasheet. These thresholds will govern the detection of camera-related faults such as overvoltage, undervoltage, communication errors, thermal shutdowns, and current-related faults.

Ensure that all thresholds (min/max) used in fault detection are derived directly from the MAX20087 datasheet. For every fault mentioned in the diagnostic requirements, CFM shall consult the relevant section in the datasheet to determine appropriate thresholds

When implementing the diagnostics for the MAX20087 power switch, the following fault types and thresholds must be handled:

- Thermal Shutdown: Reference the thermal protection section of the datasheet for shutdown temperature thresholds.
- I2C Communication Issues: Use the communication failure flags and response failure criteria from the I2C communication section.
- Input Undervoltage Lockout: Reference to the undervoltage lockout (UVLO) specification in the datasheet.
- Input Overvoltage Lockout: Reference the overvoltage lockout (OVLO) specification in the datasheet
- ISET Resistance Out of Range: Use the datasheet's current-setting resistors (ISET) section to configure detection for out-of-range resistance values. These resistances control the current limits for the channels.

- Current Out of Range: For current faults, use the ADC current reading section.

REQ-370494v1 Design Guideline: Camera Service Error Bit stream description

Purpose: This guide provides a comprehensive overview of each error code and its implications for camera diagnostics in the SW Element Camera Fault Monitor (CFM). The guide is intended to facilitate the identification and analysis of issues encountered in Low-Level Camera Device Drivers, as reported by the Camera Service via a bitstream.

The purpose of this guide is to enhance understanding of potential fault scenarios and to assist in the diagnostic process for interim problem resolution. It serves as a reference for understanding the nature of reported errors, their categorization, and expected behavior for fault handling.

Please note that this guide is not intended to serve as a specific implementation requirement. Instead, it aims to provide a unified reference for aligning the fault interpretation across different teams. Detailed diagnostic procedures and handling mechanisms should be confirmed through proper interface coordination between the Camera Service and the Camera Fault Monitor (CFM) to ensure accurate diagnostic and response to each type of error.

This provides detailed explanations of each error type generated by the Camera Service, covering its origin, possible causes, and implications for exterior camera diagnostics

- Power Deserializers: Indicates a power-related fault associated with the deserializers.
- Init Capture: Failure during the initialization of camera capture. This can indicate issues in the setup phase before streaming.
- Start Capture: Issues occurring when attempting to start camera capture
- Device Block NVSIPL Status EOF: Indicates an end-of-frame (EOF) error at the device block level, likely involving unexpected termination.
- Device Block Notif Error - Deserializer Failure: Indicates a failure detected in the Deserializer.
- Device Block Notif Error - Sensor Failure: Indicates a failure reported from the image sensor.
- Device Block Notif Error - GPIO Test Failure: Failure of the GPIO startup test, which could imply issues in signal integrity or incorrect configuration.
- Device Block Notif Error - Serializer Failure: Indicates a failure in the serializer component, suggesting a hardware problem.
- Device Block Notif Error - Internal Failure: Indicates an internal error in the device block, which could be related to hardware or firmware.
- Device Block Notif Error - Unknown: An unknown error occurred within the device block, meaning the exact cause could not be determined.
- Device Block Notif Error - Not True GPIO Interrupt: Indicates an invalid or unexpected GPIO interrupt signal.
- Pipeline Notif Error - ICP Bad Input Stream: Indicates an issue with an invalid or corrupted input stream being fed into the image capture pipeline.
- Pipeline Notif Error - ICP Capture Failure: A failure during the capture process, suggesting data may not have been captured successfully.
- Pipeline Notif Error - ICP Embedded Data Parse Failure: An error occurred while attempting to parse embedded data from the imager, which might include metadata or temperature readings.
- Pipeline Notif Error - ISP Processing Failure: Indicates a failure in the Image Signal Processor (ISP) during image processing.

- Pipeline Notif Error - ACP Processing Failure: Failure within the ACP, implying additional processing failure beyond ISP.
- Pipeline Notif Error - Deserializer Link Failure: Indicates that the link between the Deserializer and the camera sensor failed.
- Pipeline Notif Error - Internal Failure: Indicates an internal error within the pipeline, which could involve processing or hardware-related faults.
- Pipeline Notif Processing Timeout: The processing time of notifications exceeded the allowed frame time, leading to a timeout.
- Pipeline Notif Queue Timeout: A timeout occurred while attempting to retrieve notifications from the SIPL notification queue.
- Pipeline Notif Error - Unknown: An unknown error occurred within the pipeline.
- ICP Register Engine: Issues with the ICP register engine, indicating potential problems in configuring capture components.
- ISP Get Image Attributes: Failure to retrieve image attributes from the ISP, which can indicate missing or incorrect data.
- ISP Register Engine: Issues with the ISP register engine during setup or configuration.
- Read EEPROM: Indicates an error while reading from the EEPROM, which may contain sensor or calibration data.
- ICP Get Frame: Failure in retrieving a frame from the ICP, implying that captured data was lost or not available.
- ICP Parse Embedded Data - Temperature: Indicates an issue in parsing the embedded temperature data of the sensor.
- ICP Parse Embedded Data - Voltage: Indicates an error in parsing embedded voltage data from the imager.
- ICP Parse Embedded Data - SRCx: Failure to parse specific embedded data related to the SRCx channel.
- ISP Get Frame: A failure occurred while attempting to retrieve a processed frame from the ISP.
- ISP Convert YUV to RGB: An issue during the conversion of YUV format to RGB, affecting image color processing.
- Relinking: Relinking operation for the camera was initiated, typically to recover from a transient error.
- Disable Link: The link between the camera sensor and the Deserializer has been disabled.
- Enable Link: The link between the camera sensor and the Deserializer has been enabled.
- Camera Power Is Off: Indicates that the camera's power has been turned off.
- Camera Power Is Faulty: Indicates that there is a power-related issue with the camera.
- Camera Frozen Image: Indicates that the camera's output is frozen and no new frames are being captured.
- I2C Readback and Verification: Failure during the I2C readback or verification process.
- Image Timestamp Deviation Exceeded: Indicates that the timestamp deviation for the captured image exceeds allowable limits.
- Image Timestamp Integrity False: Indicates that the integrity check for image timestamps has failed.

- Camera Temperature Out of Range: Indicates that the camera temperature is outside the acceptable operating range.
- Pipeline Notif Error - CDI Set Sensor Control Failure: A failure occurred when attempting to set sensor control through CDI.
- Parse Image Metadata: Indicates an error while parsing metadata from the captured image.
- Device Block Notif Error - Get Deserializer Error Info Failure: Failure to retrieve error information from the Deserializer.
- Device Block Notif Error - Get Module Error Info Failure: Failure to retrieve detailed error information from a specific module in the device block.
- Device Block Notif Error - Interrupt Localization Failure: Indicates a failure in localizing the source of an interrupt.
- Device Block Notif Error - Interrupt Localization Timeout: Indicates a timeout during the process of localizing the interrupt source.
- Unknown Error: An unspecified error occurred, and its nature could not be determined.

REQ-413301v1 CFM - Camera recovery publication interface

Purpose: Provide downstream consumers (e.g. the Diagnostic Watchdog Monitor) receive, for each camera, a clear indication of when any permanent camera fault (not just init failures) has occurred and which recovery action is needed to restore functionality.

Rationale: By including any permanent fault (rather than only init failures), CFM capture all scenarios where a reset or suspend/resume is the

If there are new camera interfaces simply add their own Camera Recovery Needs property following this pattern, with no changes to the requirement.

SWE Camera Fault Monitor (CFM) shall monitor all matured, non-transient camera faults across every supported camera interface (e.g. forward-looking wide-angle, fisheye short-range surround-view) and determine, for each affected camera, the appropriate recovery action.

CFM shall then expose a subscribable *Camera Recovery Needs* property on each camera interface in a structured list pairing each camera identifier with its required recovery action.

REQ-447154v2 Design Guideline: CFM - Snapshot information

Fault Identifier Provided By Camera Service

ID	Category	Description
0x00	kNoFault	Indicates that there are no faults detected.
0x01	kBadVideoData	Video input from the cameras is bad.
0x02	kCaptureTimeoutError	Waiting on captured frame timed out.
0x03	kDeserializerElectricalFault	Voltage errors reported from the deserializer.
0x04	kDeserializerFault	Deserializer fault.
0x05	kDeserializerGlobalFault	Global faults reported from the deserializer.
0x06	kDeserializerLinkError	Link errors reported from the deserializer.
0x07	kDeserializerPipelineFault	Pipeline faults reported from the deserializer.
0x08	kEepromDataIntegrityCheckFault	EEPROM data integrity check failed, indicating that the contents are faulty, or that it's an unsupported camera module sample revision.
0x09	kEepromReadFault	Failed to read the EEPROM.
0x10	kFsyncError	Fsync error.
0x11	kInitializationAndStartupFault	Failed to initialize or start camera.

0x12	kInvalidImageMetadata	Issues in the image metadata.
0x13	kLinkLockFault	Communication with camera module, sensor, serializer, or deserializer failed, indicating a link lock fault.
0x14	kModuleElectricalFault	Voltage errors in any of the camera components.
0x15	kSafetyMechanismCheckFault	Safety mechanism check fault. Failed to verify that safety mechanisms are enabled.
0x16	kSensorFault	Sensor fault.
0x17	kSensorGpioTestFault	Sensor GPIO test fault. Failed to verify that sensor error reporting is enabled.
0x18	kSerializerFault	Serializer fault.
0x19	kSerializerGpioTestFault	Serializer GPIO test fault. Failed to verify that serializer error reporting is enabled.
0x20	kSystemFault	Platform faults that are not reported by degradation manager.
0x21	kTemperatureOutOfRange	Temperature out of range faults.
0x22	kUnknownDriverErrors	Camera driver reported unknown fault.
0x23	kVehicleTimeFault	Faults related to vehicle time.